

BONSA ADUGNA

Software Engineer/ML Engineer

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EDUCATION

B.Sc. in Computer Science and Engineering

05/2022 – 06/2026

Adama Science and Technology University | CGPA: 3.63 / 4

Adama, Ethiopia

4.9 Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Information Retrieval, Artificial Intelligence, Computer Networks, Computer Architecture, Operating Systems, Concurrency, Object-Oriented Design, Advanced Java, C++ Programming, Requirements Engineering.

Real-Time Backend Systems Program

07/2025 – 05/2026

EPIC Institute of Technology

Newtown, Pennsylvania

- Designed and implemented real-time backend architectures using distributed systems, Raft consensus, and fault-tolerant design.
- Built large-scale data pipelines with **AWS** and **Docker**; applied time series analysis for real-time insights.

Competitive Programming and Software Development

11/2023 – 09/2024

Africa To Silicon Valley

Adama, Ethiopia

- Solved **800+** algorithmic problems on LeetCode and Codeforces.

PROFESSIONAL EXPERIENCE

Robotic Games Inc. (Funded by Founders)

08/2025 – Present

Software Engineer

San Francisco, CA

- Applied **SimHash-based deduplication** with adaptive Hamming thresholds, reducing redundant analyses by **60-80%** using Python and Redis.
- Created a content-matching pipeline on AWS (FastAPI, PostgreSQL), lowering API costs by **60%** and reducing median latency from **2.3 s to 0.8 s** through Dockerized services and optimized indexing.
- Developed a browser extension (React, WebExtensions API) that filters unwanted web content in real-time, cutting scan time to under **20 ms**.
- Engineered a low-latency backend (**< 370 ms**) with FastAPI and Redis, integrating an on-device privacy layer for secure personalization.

Africa to Silicon Valley

01/2025 – Present

Head of Education

Adama, Ethiopia

- Mentored **60+** students in software development and data structures; organized **19 contests** and authored solutions, resulting in **22 internships and job placements**.
- Recognized as **Best Lecturer** among eight educators for instructional quality and student engagement.

Eskalate

Project Lead (Sparck, AI Social Media Platform)

12/2024 – 08/2025

Palo Alto, CA

- Led a **15-member team** to develop **Sparck**, a social media management platform for creation, scheduling, multi-channel publishing, and analytics with **98.9% uptime**.
- Created **50+ RESTful APIs** with OAuth2/JWT authentication, rate limiting, and cursor pagination; improved observability with structured logs, traces, and SLOs.
- Reduced feature lead time by **30%** via query optimization, Redis caching, and connection tuning.
- Containerized services with Docker and automated CI/CD pipelines using GitHub Actions

Software Engineering Intern

07/2024 – 09/2024

Palo Alto, CA

- Developed **socket-based chat features** supporting **500+ concurrent users** with zero-downtime releases.
- Improved code quality by driving prototyping, testing, and reviews, increasing test coverage from **55% to 78%**.

SKILLS

Languages: Python, C, C++, Java, JavaScript, Dart, Rust, SQL | **Frameworks & Libraries:** Django REST Framework, FastAPI, Next.js, React, Flutter, Flask, Express.js | **ML/AI:** PyTorch, scikit-learn, Hugging Face Transformers, RAG, Vector Search (FAISS, Pinecone), NumPy, Pandas | **Databases:** MySQL, MongoDB, PostgreSQL, SQLite | **Cloud / Tools:** AWS, GCP, Docker, Git & GitHub, CI/CD, Linux

PROJECTS

Career Pilot — AI-Powered Career Guidance Platform

- Designed a responsive frontend with Next.js and Framer Motion; integrated FastAPI microservices to serve **280+ monthly active users**.
- Established a robust authentication system using OAuth 2.0 and role-based access control, resulting in zero incidents of unauthorized access and meeting stringent compliance requirements.

QuizMe – Adaptive AI-Driven Quiz Platform

- Engineered a system with **RAG pipelines, vector search, and prompt orchestration**, processing over **80 quizzes per day**.
- Delivered a prototype using Python, Django, LangChain, and FAISS/Pinecone; ranked **Top 30** among **1,000+ teams** nationally.

ACHIEVEMENTS AND AWARDS

- African and Arab Collegiate Programming Championship (ACPC) Finalist** | **Expert Rating on Codeforces Top 5% globally** | **A2SV Generative AI for Africa Hackathon Semi-Finalist**